

Divergence-Regularized Federated GANs for Effective Cyber-Attack Detection on Non-IID and Unlabeled Edge Activity Data

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Abstract—Edge computing enables real-time Internet of Things data processing by bringing computation closer to data sources, but its distributed architecture creates cybersecurity vulnerabilities requiring privacy-preserving attack detection mechanisms capable of handling heterogeneous data distributions. This article proposes federated generative adversarial divergence (FedGAD), a plug-and-play modular framework that enhances existing federated learning methods through Jacobian-based regularization and dynamic complexity-aware weighting to address cyber-attack detection in non-independent and identically distributed (IID) and unlabeled edge data environments. Unlike existing approaches suffering from mode collapse and training instability, FedGAD maintains statistical consistency across distributed nodes through gradient-based stability mechanisms, supported by rigorous theoretical analysis establishing convergence guarantees and mode coverage properties. We conduct comprehensive experiments comparing FedGAD against four federated generative learning baselines: federated trustworthy (FedTrust), anomaly detection generative adversarial network (ADGAN), federated generative adversarial network for intrusion detection system (FedGAN-IDS), and federated temporal sequential recurrent generative network (FedTSRGNet) and four regularization-based methods: federated averaging (FedAvg), federated proximal (FedProx), learning with collaborative aggregation method (LeCam), and Jensen Shannon (JS) Divergence on telemetry data of networks - internet of things (ToN_IoT) and Communications Security Establishment in Canadian Institute for Cybersecurity - Intrusion Detection System (CSE_CIC_IDS) datasets, demonstrating FedGAD's superiority with accuracy improvements up to 3.5%, achieving 100% mode coverage compared to 25%

for baseline methods while maintaining computational efficiency for resource-constrained edge deployments.

Index Terms—Cyber-attack detection, edge computing (EC), federated learning (FL), generative adversarial networks, Jacobian regularization, non-independent and identically distributed (IID) data.

I. INTRODUCTION

EDGE computing (EC) processes data locally at edge servers, enabling real-time processing for Internet of Things (IoT) applications [1]. This reduces latency, conserves bandwidth, and enhances privacy for autonomous vehicles, healthcare systems, smart grids, and remote monitoring. The global EC market is projected to reach USD 155.90 billion by 2030¹ or USD 5132.29 billion by 2034.² However, EC's distributed architecture is susceptible to denial of service (DoS) attacks, man-in-the-middle attacks, and advanced persistent threats [2], [3], necessitating intelligent detection mechanisms for EC environments.

Traditional centralized intrusion detection system (IDS) [4] are inadequate for EC, requiring centralized data collection that violates privacy and creates scalability bottlenecks [5], while struggling with high-dimensional, unlabeled, non-independent and identically distributed (IID) data from diverse edge devices [6]. Federated learning (FL) addresses these limitations by enabling collaborative model training across edge nodes without data sharing [7]. Using federated averaging (FedAvg) [8], FL aggregates local models to create a global model. However, classical FL performs poorly on non-IID and unlabeled data, leading to suboptimal detection performance. Recent machine learning advances show promise in automating attack detection [9], but fail under unlabeled and non-IID conditions. To overcome these challenges, researchers have integrated generative adversarial networks (GANs) with FL [10], [11], where generators model data distributions and discriminators distinguish real from synthetic samples to compensate for data imbalances.

Traditional centralized detection systems face the following three fundamental challenges in ECs:

¹[Online]. Available: <https://www.grandviewresearch.com/press-release/global-edge-computing-market>

²[Online]. Available: <https://www.precedenceresearch.com/edge-computing-market>

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Code is available online at: <https://github.com/TII-25-8978/FedGAD>. Digital Object Identifier 10.1109/TII.2026.3666264

- 1) privacy violations through mandatory data centralization;
- 2) scalability bottlenecks due to bandwidth constraints;
- 3) inability to handle unlabeled and heterogeneous non-IID data from diverse edge devices.

While FL addresses privacy, existing FL-GAN approaches suffer from mode collapse under non-IID conditions, failing to capture attack pattern diversity, especially for rare sophisticated attacks. This worsens with unlabeled data common in edge deployments.

Despite this integration, FL-GAN approaches face fundamental issues with non-IID data. Heterogeneous client distributions prevent GANs from learning unified representations of global data diversity. This leads to mode collapse, where generators produce limited output varieties, failing to capture complete data distributions [12]. The problem intensifies in federated settings due to the absence of global labels. This can cause training instability and poor generalization [13]. Existing solutions like mini-batch discrimination and feature matching introduce computational overhead or require labels, making them unsuitable for resource-constrained edge environments [14].

To overcome these challenges, this article proposes federated generative adversarial divergence (FedGAD), which advances beyond existing FL-GAN frameworks by directly addressing the three fundamental limitations in ECs by preserving privacy through fully decentralized training. This ensures scalability via efficient model aggregation without raw data transmission and handling unlabeled non-IID data through unsupervised adversarial learning with heterogeneity-aware mechanisms. We achieve these solutions through the following four key technical contributions.

- 1) We introduce a differential geometric regularization technique that directly regularizes the generator's Jacobian matrix to prevent mode collapse and enhance attack pattern diversity. Unlike prior works [15] that apply Jacobian regularization to supervised discriminative models, we pioneer its integration within unsupervised federated adversarial training, demonstrating its effectiveness in maintaining statistical consistency between heterogeneous local and global distributions without requiring labeled data.
- 2) Unlike the fixed regularization coefficients in federated temporal sequential recurrent generative network (FedT-SRGNet) [16] and federated generative adversarial network for intrusion detection system (FedGAN-IDS) [11], we propose a dynamic weighting mechanism that adapts regularization strength to local data complexity at each edge node. This heterogeneity-aware approach effectively mitigates the non-IID induced mode collapse that plagues existing static methods.
- 3) We provide rigorous mathematical analysis establishing the following:
 - a) Lipschitz continuity guarantees through gradient penalty stability (Theorem 4.1);
 - b) almost-sure convergence to stationary points under federated settings (Theorem 4.2);
 - c) enhanced mode coverage proofs (Theorem 4.3).

This theoretical foundation distinguishes our work from empirically driven approaches.

- 4) We design FedGAD as a modular framework that can be seamlessly integrated with existing federated methods, enhancing their performance without architectural modifications. To validate this extensibility, we conduct comprehensive experiments integrating FedGAD with four state-of-the-art approaches, such as federated trustworthy (FedTrust) [17], anomaly detection generative adversarial network (ADGAN) [18], FedGAN-IDS [11], and FedT-SRGNet [16], demonstrating consistent performance improvements over their original implementations across multiple evaluation metrics and heterogeneity scenarios.

The rest of this article is organized as follows. Section II provides a critical analysis of the literature. We then present our proposed FedGAD approach in Section III. Section IV derives theoretical convergence guarantees. In Section V, we provide the experimental setup and results on real-world public datasets. Finally, Section VI concludes this article.

II. RELATED WORK

Recent years have seen significant efforts to improve cyber-attack detection in EC environments using FL- and GAN-based approaches. These techniques aim to address the limitations of centralized, rule-based systems in decentralized, privacy-sensitive, and heterogeneous settings. This section critically examines the existing literature, identifies its fundamental limitations, and positions our contributions within the current research landscape.

A. FL for Intrusion Detection

Several studies have proposed FL-based IDS solutions tailored for EC. Rajarajan et al. [19] introduced an explainable federated framework for secure and privacy-preserving IDS in vehicular edge networks, highlighting the importance of interpretability under adversarial conditions. Similarly, Friha et al. [20] proposed a decentralized FL-based IDS for edge devices, and Chen et al. [21] presented a multimodal privacy-preserving FL approach for critical IoT systems. Recent advances include a comprehensive framework [22] that has been developed for an optimal FL-based intrusion detection specifically optimized for IoT environments. However, these approaches share a critical limitation: they rely on the standard FedAvg aggregation [8], which assumes relatively homogeneous data distributions. Under severe non-IID conditions common in EC, where different nodes observe vastly different attack patterns, model updates can diverge significantly, leading to biased global models and degraded detection accuracy [23], [24]. Furthermore, these methods require substantial labeled data for supervised training, which is often unavailable or expensive to obtain in dynamic edge environments.

B. Federated GANs and Mode Collapse

To handle data distribution challenges, researchers have integrated GANs into FL frameworks [17]. Xiong et al. [10] proposed a distributed generative model for heterogeneous data using local generators, while Tabassum et al. [11] introduced FedGAN-IDS, a privacy-preserving framework

combining GANs with FL for intrusion detection. Recent survey work [25] analyzing FL-based intrusion detection in medical IoT highlighted FLIDS systems achieving over 99% detection accuracy through GAN-FL integration.

Despite these promising results, FedGAN-IDS suffers from fundamental mode collapse under non-IID conditions, where local generators training on skewed distributions fail to learn comprehensive global data representations, producing synthetic samples that cluster around dominant modes while missing rare attack types particularly severe in cybersecurity where rare sophisticated attacks (e.g., advanced persistent threats) are precisely the patterns that must be detected. Various strategies have been explored to mitigate mode collapse, but face critical limitations for edge deployments. Mini-batch discrimination [14] introduces significant computational overhead unacceptable for resource-constrained edge nodes. Feature matching requires careful hyperparameter tuning and can destabilize in federated settings with heterogeneous hardware. Unrolled GANs [26] improve stability but incur $3\text{--}5\times$ higher gradient computation, making them impractical for edge devices. Spectral normalization [27] constrains discriminator Lipschitz constants but lacks adaptive mechanisms for varying data distributions across clients.

C. FL With Regularization

Federated proximal (FedProx) [23] addresses heterogeneity through proximal terms that limit local model deviation from the global model. While effective for supervised learning with labeled data, FedProx provides no mechanism for handling mode collapse in generative models; its proximal regularization operates on model parameters rather than data manifolds. SCAFFOLD [28] uses control variates to correct client drift but similarly assumes supervised objectives and fails to address generative training instabilities.

D. Regularization Techniques in Federated GANs

Regularization has emerged as a critical mechanism for stabilizing federated GAN training, though existing approaches remain limited in simultaneously addressing mode collapse, convergence guarantees, and edge deployment requirements. Gradient penalty methods, such as WGAN-GP [12] enforce Lipschitz continuity but are poorly suited to federated settings. Federated GAN-based approaches, such as FedTrust [17], ADGAN [18], and FedGAN-IDS [11]. More recently, FedTSRGNet [16] combines bidirectional long short-term memory (Bi-LSTM), temporal convolutional networks (TCN), and GANs for cyber-attack detection. However, it employs only static regularization without adaptation to local data complexity, providing no theoretical analysis of convergence properties or mode coverage guarantees. Our work fundamentally differs in the following:

- 1) introducing adaptive Jacobian regularization that dynamically adjusts to local distribution heterogeneity;
- 2) providing rigorous theoretical guarantees for convergence and mode coverage;
- 3) demonstrating through ablation studies that our regularization mechanism specifically addresses mode collapse

TABLE I
COMPARISON OF FEDGAD WITH RELATED WORKS

Method	Adaptive Reg.	Convergence Guarantee	Mode Coverage	Unlabeled Data	Edge Efficient
FedAvg [8]	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$
FedProx [23]	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$
FedGAN-IDS [11]	$\times$	$\times$	$\times$	$\checkmark$	$\times$
FedTSRGNet [16]	$\times$	$\times$	Partial	$\checkmark$	$\times$
FedGAD (Ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

The bold values denote the highest performance metrics.

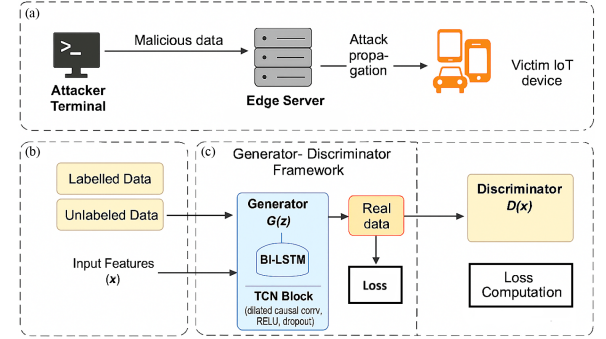


Fig. 1. Overview of the proposed FedGAD pipeline. (a) Attacker injects malicious data into the edge server that communicates with edge/IoT devices. (b) Both labeled and unlabeled data are collected for training. (c) Generator combines Bi-LSTM and TCN blocks to capture temporal dependencies, while the discriminator distinguishes real and fake samples. The overall loss combines adversarial and reconstruction objectives.

by achieving 100% mode coverage versus 0% missing modes compared to FedTSRGNet's incomplete coverage.

Table I provides a systematic comparison of FedGAD and representative prior works. Unlike existing approaches, FedGAD combines the following:

- 1) distribution-aware adaptive regularization;
- 2) theoretical convergence guarantees for nonconvex federated adversarial training;
- 3) mode collapse mitigation without labeled data requirements;
- 4) computational efficiency suitable for edge deployment.

III. PROPOSED METHODOLOGY

This section presents the proposed FedGAD framework for cyber-attack detection in heterogeneous and unlabeled edge data environments. Building upon the FedTSRGNet baseline, FedGAD introduces Jacobian-based regularization and adaptive weighting mechanisms to overcome instability and mode collapse commonly encountered in federated generative adversarial networks. We provide detailed architectural descriptions, algorithmic procedures with step-by-step explanations, and comprehensive complexity analysis. The overall architecture and federated training pipeline are illustrated in Fig. 1.

A. FedTSRGNet and Motivating Limitations

FedTSRGNet integrates a temporal sequence recurrent network with a GAN in an FL setting for cyber-attack detection. As shown in Fig. 1, it comprises a generator G and a discriminator D . The generator processes input data through an embedding layer, followed by a Bi-LSTM encoder with TCN blocks using rectified linear unit (ReLU) activation, weight normalization,

and dilated convolutions, along with dropout and sequentially synthesis data (SSD) modules for sequential synthesis. The discriminator evaluates both real (labeled or unlabeled) and generated samples, applying a regularizer λ to guide training. The model leverages multiple loss functions ($\mathcal{L}_G$, $\mathcal{L}_D$, $\mathcal{L}_{SSD}$, $\mathcal{L}_{reg}$, $\mathcal{L}_{total}$) to learn and generate synthetic attack patterns effectively. Despite the strengths of FedTSRGNet, it suffers from performance degradation in the presence of non-IID and unlabeled data, which are inherent in real-world edge environments. In particular, the generator in the GAN component tends to experience mode collapse, generating a narrow set of outputs that fail to represent the full diversity of cyber-attack patterns. This problem is further exacerbated by the absence of global labels and highly skewed local data distributions across clients, resulting in unstable training and suboptimal global model performance.

B. Problem Formulation and Design Objective

Let $i = \{1, 2, \dots, N\}$ be the set of participating edge nodes, where each node $i \in \mathcal{N}$ holds a local dataset $\mathcal{D}_i = \{x_j^{(i)}\}_{j=1}^{n_i}$ drawn from a unique distribution $p_i(x)$. Due to variations in infrastructure, attack types, and local contexts, the divergence between any two local distributions is typically large, formally expressed as $D_{KL}(p_i(x)|p_j(x)) \gg 0$ for $i \neq j$, where D_{KL} denotes the Kullback–Leibler divergence [29]. This divergence results in non-IID conditions that significantly hinder the effectiveness of classical FL techniques and exacerbate instability in adversarial training. The original FedTSRGNet employs a generator network $G_\theta: \mathbb{R}^d \rightarrow \mathbb{R}^m$ parameterized by θ , which generates synthetic data from a latent dimension space d , and a discriminator network $D_\phi: \mathbb{R}^m \rightarrow [0, 1]$ parameterized by ϕ , which attempts to differentiate between real and generated data in the dimension feature space m . Under non-IID conditions, adversarial training becomes unstable, and the generator tends to collapse, producing only a few repetitive patterns. To address this, we propose an enhanced architecture called FedGAD that incorporates geometric regularization and an adaptive training protocol.

C. FedGAD Architecture

FedGAD extends FedTSRGNet through the following three key architectural innovations.

- 1) *Generator architecture*: The generator G_θ employs a hybrid architecture combining embedding layers (dimension $d_{emb} = 128$), Bi-LSTM layers (hidden size $h_{lstm} = 256$), and TCN blocks with exponentially increasing dilation rates $\{1, 2, 4, 8\}$. This design captures both long-term temporal dependencies through Bi-LSTM and fine-grained local patterns through dilated convolutions. The SSD (sequential synthesis data) module generates synthetic samples through attention mechanisms, enabling focus on relevant attack signatures.
- 2) *Discriminator with Jacobian regularization*: Unlike standard discriminators, our discriminator D_ϕ incorporates gradient computation layers that enable Jacobian penalty calculation. The architecture consists of the following:
 - a) input processing layers (Conv1D, BatchNorm);
 - b) feature extraction blocks;

Algorithm 1: Model Training in FedGAD (Client-Side).

Require: Local dataset $\mathcal{D}_i$, local epochs E , learning rates η_G, η_D , global parameters $\theta^{(t)}, \phi^{(t)}$, base regularization λ_{base}

Ensure: Updated local parameters $\theta_i^{(t+1)}, \phi_i^{(t+1)}$, complexity measure $\mathcal{C}_i$

- 1: // Phase 1: Initialization
- 2: $\theta_i^{(t)} \leftarrow \theta^{(t)}, \phi_i^{(t)} \leftarrow \phi^{(t)}$ ▷ Download global parameters
- 3: // Phase 2: Compute local data complexity
- 4: Compute local distribution complexity $\mathcal{C}_i$ via kernel density estimation:
- 5: $\mathcal{C}_i = \frac{1}{|\mathcal{D}_i|} \sum_{x \in \mathcal{D}_i} \|\nabla_x \log \hat{p}_i(x)\|_2^2$
- 6: // Phase 3: Receive adaptive weight from server
- 7: Receive $\lambda_{J,i}$ from server based on (2)
- 8: **for** epoch $e = 1$ to E **do**
- 9: // Phase 4: Data sampling
- 10: Sample mini-batch $\mathcal{B}_i \subset \mathcal{D}_i$ of real attack patterns
- 11: Sample latent vectors $\mathcal{Z} \sim \mathcal{N}(0, I_d)$ from latent distribution
- 12: // Phase 5: Generator update
- 13: Generate fake samples: $\tilde{x} = G_{\theta_i^{(t)}}(z)$ for $z \in \mathcal{Z}$
- 14: Compute generator loss:
 $\mathcal{L}_G = \mathbb{E}_z[\log(1 - D_{\phi_i^{(t)}}(G_{\theta_i^{(t)}}(z)))]$
- 15: Update generator: $\theta_i^{(t)} \leftarrow \theta_i^{(t)} - \eta_G \nabla_{\theta} \mathcal{L}_G$
- 16: // Phase 6: Discriminator update with Jacobian penalty
- 17: Compute adversarial loss:
- 18: $\mathcal{L}_D^{orig} = -\mathbb{E}_{x \in \mathcal{B}_i}[\log D_{\phi_i}(x)] - \mathbb{E}_z[\log(1 - D_{\phi_i}(G_{\theta_i}(z)))]$
- 19: Compute Jacobian penalty:
 $\mathcal{R}_J = \frac{1}{|\mathcal{B}_i|} \sum_{x \in \mathcal{B}_i} \|\nabla_x D_{\phi_i}(x)\|_2^2$
- 20: Update discriminator:
 $\phi_i^{(t)} \leftarrow \phi_i^{(t)} - \eta_D \nabla_{\phi}[\mathcal{L}_D^{orig} + \lambda_{J,i} \mathcal{R}_J]$
- 21: **end for**
- 22: // Phase 7: Upload to server
- 23: Upload $\theta_i^{(t+1)}, \phi_i^{(t+1)}$, and $\mathcal{C}_i$ to central server
- 24: **return** Updated parameters $\theta_i^{(t+1)}, \phi_i^{(t+1)}$, complexity $\mathcal{C}_i$

- c) gradient computation module for $\nabla_x D_\phi(x)$;
- d) classification head outputting probability scores in $[0, 1]$.

This design ensures efficient gradient flow for both adversarial training and regularization.

- 3) *Adaptive weighting module*: Each edge node computes local complexity $\mathcal{C}_i$ using kernel density estimation over mini-batches, approximating $\|\nabla_x \log p_i(x)\|_2^2$. The central server maintains running statistics ($\bar{\mathcal{C}}, \sigma_{\mathcal{C}}$) and computes node-specific weights via (2), enabling heterogeneity-aware regularization strength adjustment.

D. FedGAD Regularized Training

FedGAD expands the FedTSRGNet framework by integrating Jacobian-based regularization into the discriminator to improve

Algorithm 2: Model Aggregation in FedGAD (Server-Side).

Require: Edge nodes $\mathcal{N} = \{1, \dots, N\}$, communication rounds T , base regularization λ_{base} , scaling parameter α , convergence threshold ϵ

Ensure: Global generator G_{θ^*} and discriminator D_{ϕ^*}

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1: // Phase 1: Initialization
2: Initialize  $\theta^{(0)}, \phi^{(0)}$  randomly
3: Initialize  $\bar{C} \leftarrow 0, \sigma_C \leftarrow 1$ 
4: for round  $t = 0$  to  $T - 1$  do
5:   // Phase 2: Broadcast to clients
6:   Broadcast  $\theta^{(t)}, \phi^{(t)}$  to all nodes  $i \in \mathcal{N}$ 
7:   // Phase 3: Compute and broadcast adaptive weights
8:   for each node  $i \in \mathcal{N}$  do
9:     Compute adaptive weight using (2):
10:     $\lambda_{J,i} = \lambda_{base} \cdot (1 + \alpha \cdot \frac{C_i - \bar{C}}{\sigma_C})$ 
11:    Send  $\lambda_{J,i}$  to node  $i$ 
12:   end for
13:   // Phase 4: Wait for client updates
14:   Execute Algorithm 1 at each client
15:   Receive  $\{\theta_i^{(t+1)}, \phi_i^{(t+1)}, C_i\}$  from all  $i \in \mathcal{N}$ 
16:   // Phase 5: Update global complexity statistics
17:    $\bar{C} \leftarrow \frac{1}{N} \sum_{i=1}^N C_i$ 
18:    $\sigma_C \leftarrow \sqrt{\frac{1}{N} \sum_{i=1}^N (C_i - \bar{C})^2}$ 
19:   // Phase 6: Weighted aggregation
20:    $\theta^{(t+1)} = \sum_{i=1}^N \frac{|D_i|}{|\mathcal{D}|} \theta_i^{(t+1)}$ 
21:    $\phi^{(t+1)} = \sum_{i=1}^N \frac{|D_i|}{|\mathcal{D}|} \phi_i^{(t+1)}$ 
22:   // Phase 7: Convergence check
23:   if  $\|\theta^{(t+1)} - \theta^{(t)}\|_2 + \|\phi^{(t+1)} - \phi^{(t)}\|_2 < \epsilon$  then
24:     break  $\triangleright$  Convergence achieved
25:   end if
26: end for
27: return  $G_{\theta^{(T)}}, D_{\phi^{(T)}}$ 

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training stability and prevent mode collapse. During each communication round t , edge node i downloads the current global model parameters $\theta^{(t)}$ and $\phi^{(t)}$ from the central server. The node then computes a local distribution complexity measure C_i using kernel density estimation, which captures the inherent heterogeneity of its local data distribution $p_i(x)$. During local training, the generator continues to update its parameters using the standard adversarial loss. In contrast, the discriminator incorporates a Jacobian regularization term defined as

$$\mathcal{R}_J(x; \phi) = \lambda_J \mathbb{E}_{x \sim p_{data}} [\|\nabla_x D_\phi(x)\|_2^2] \quad (1)$$

where $\lambda_J > 0$ is the regularization coefficient. This term penalizes large gradients and encourages Lipschitz continuity, thereby preventing sharp decision boundaries and supporting smoother transitions between data modes. To accommodate varying degrees of heterogeneity across nodes, we introduce an adaptive weighting scheme where the regularization coefficient for each node is computed as

$$\lambda_{J,i} = \lambda_{base} \cdot \left(1 + \alpha \cdot \frac{C_i - \bar{C}}{\sigma_C}\right) \quad (2)$$

where $C_i = \frac{1}{|\mathcal{D}_i|} \sum_{x \in \mathcal{D}_i} \|\nabla_x \log p_i(x)\|_2^2$ with $\bar{C}$ and σ_C represent the global mean and standard deviation of local complexity scores, respectively. This dynamic adjustment ensures that edge nodes with more complex or diverse data distributions receive proportionally stronger regularization. Now, the generator parameters $\theta_i^{(t+1)}$ are updated via

$$\theta_i^{(t+1)} = \theta_i^{(t)} - \eta_G \nabla_{\theta} \mathbb{E}_{z \sim p_z} [\log(1 - D_{\phi^{(t)}}(G_{\theta^{(t)}}(z)))] \quad (3)$$

and the regularized discriminator parameters $\phi_i^{(t+1)}$ are updated via

$$[b]\phi_i^{(t+1)} = \phi_i^{(t)} - \eta_D \nabla_{\phi} [\mathcal{L}_D^{\text{orig}}(\phi_i^{(t)}) + \lambda_{J,i} \mathbb{E}_{x \sim \mathcal{D}_i} [\|\nabla_x D_{\phi^{(t)}}(x)\|_2^2]] \quad (4)$$

where η_G and η_D are the learning rates for the generator and discriminator, respectively, and $\mathcal{L}_D^{\text{orig}}(\phi) = -\mathbb{E}_{x \sim p_{data}} [\log D_\phi(x)] - \mathbb{E}_{z \sim p_z} [\log(1 - D_\phi(G_\theta(z)))]$ represents the standard adversarial loss. After local training, each node transmits its updated parameters and complexity measure to the central server. The server performs weighted aggregation based on the size of each node's dataset

$$\theta^{(t+1)} = \sum_{i=1}^N \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \theta_i^{(t+1)}, \quad \phi^{(t+1)} = \sum_{i=1}^N \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \phi_i^{(t+1)} \quad (5)$$

where $|\mathcal{D}| = \sum_{i=1}^N |\mathcal{D}_i|$ is the total number of samples across all nodes. The server also updates the global statistics of the complexity measures to be used in the next communication round. The training process, as illustrated in Algorithm 1, and the aggregation procedure, as illustrated in Algorithm 2, continue iteratively until the convergence criteria are satisfied. These criteria include stabilization of generator and discriminator loss values and minimal improvements in model performance across successive rounds. This convergence assessment helps avoid over training and reduces unnecessary communication overhead, making the approach suitable for real-time or resource-constrained edge deployments.

E. Complexity Analysis

In this section, we provide detailed computational and communication complexity analysis.

1) **Computational Complexity:** We analyze the computational overhead of FedGAD at both client and server sides, demonstrating that while the Jacobian penalty doubles discriminator computation, the overall complexity remains tractable for edge deployments with mini-batch approximations reducing quadratic costs to linear operations.

Client-side computation: Each client performs the following operations per training round.

- 1) *Complexity estimation:* Computing C_i via kernel density estimation requires $O(n_i^2 \cdot m)$ operations for n_i local samples and m -dimensional features. In practice, we use mini-batch approximation, reducing this to $O(B \cdot n_i \cdot m)$ where B is the batch size.
- 2) *Generator update:* Forward pass through Bi-LSTM and TCN layers costs $O(E \cdot B \cdot (h_{\text{lstm}}^2 + k \cdot h_{\text{tcn}}))$ where E is

local epochs, h_{lstm} is LSTM hidden size, k is TCN kernel size, and h_{tcn} is TCN hidden channels. Backward pass has identical complexity.

- 3) *Discriminator update*: Standard forward–backward pass costs $O(E \cdot B \cdot |\phi| \cdot m)$ where $|\phi|$ is discriminator parameter count. Jacobian penalty computation adds $O(E \cdot B \cdot |\phi| \cdot m)$ for gradient calculation via automatic differentiation.
- 4) *Total client complexity*: $O(B \cdot n_i \cdot m + E \cdot B \cdot (|\theta| + 2|\phi| \cdot m))$ where $|\theta|$ represents generator parameters.

Server-side computation: The server performs computing $\bar{C}$ and σ_C costs $O(N)$ for N clients. Computing $\lambda_{J,i}$ for all clients costs $O(N)$ for weight calculation. Then, weighted averaging of parameters costs $O(N \cdot (|\theta| + |\phi|))$ calculated for model aggregation. Total server complexity is $O(N \cdot (|\theta| + |\phi|))$ per round.

2) *Communication Complexity*: Communication costs dominate FL overhead by each client exchanges $(|\theta| + |\phi|)$ parameters and a scalar complexity value, totaling $O(|\theta| + |\phi|)$ per client per round. For T rounds and N clients: $O(T \cdot N \cdot (|\theta| + |\phi|))$. FedGAD maintains identical communication complexity to FedTSRGNet and FedProx since regularization is computed locally. This contrasts with approaches requiring auxiliary model exchanges. The $2\times$ computational overhead in discriminator updates is offset by: 1) faster convergence, reducing required rounds T by 30%–40% (see Fig. 5) and 2) improved mode coverage, eliminating the need for post hoc augmentation. Empirically, FedGAD achieves convergence in 50 epochs versus 80–100 for baselines, yielding net computational savings.

IV. THEORETICAL RESULT ANALYSIS

This section provides formal guarantees and analytical insights into the stability, convergence, and effectiveness of the proposed FedGAD framework. We demonstrate that the Jacobian-based regularization enforces Lipschitz continuity in the discriminator, stabilizing adversarial training and mitigating mode collapse. We further prove that, under standard assumptions on gradient boundedness and learning rates, FedGAD converges to a stationary point of the regularized mini–max objective. Finally, we analyze the computational and communication complexities and establish theoretical support for FedGAD’s ability to enhance mode coverage under non-IID and unlabeled data conditions.

Assumptions: Our theoretical analysis relies on the following standard assumptions from FL and optimization literature [23], [28].

- 1) *Bounded gradients*: There exists $G_{\max}$ such that $\|\nabla_{\theta} \mathcal{L}_G(\theta)\|_2 \leq G_{\max}$ and $\|\nabla_{\phi} \mathcal{L}_D(\phi)\|_2 \leq G_{\max}$ for all θ, ϕ .
- 2) *Bounded variance*: Local gradient variance is bounded: $\mathbb{E}\|\nabla \mathcal{L}_i - \nabla \mathcal{L}\|_2^2 \leq \sigma^2$ for all clients i .
- 3) *Smoothness*: Loss functions are L -smooth: $\|\nabla \mathcal{L}(w_1) - \nabla \mathcal{L}(w_2)\|_2 \leq L\|w_1 - w_2\|_2$.
- 4) *Learning rate conditions*: Learning rates satisfy $\sum_{t=1}^{\infty} \eta_t = \infty$ and $\sum_{t=1}^{\infty} \eta_t^2 < \infty$ (e.g., $\eta_t = \eta_0/\sqrt{t}$).

A. Gradient Penalty Stability

Theorem IV.1 (Gradient Penalty Stability): The Jacobian regularization term $\mathcal{R}_J(x; \phi)$ promotes Lipschitz continuity in the discriminator function, thereby stabilizing the adversarial training dynamics.

Proof: Consider the discriminator function $D_{\phi} : \mathbb{R}^m \rightarrow [0, 1]$. For any two points $x_1, x_2 \in \mathbb{R}^m$, the mean value theorem guarantees the existence of a point $\xi \in [x_1, x_2]$ such that

$$D_{\phi}(x_2) - D_{\phi}(x_1) = \nabla_x D_{\phi}(\xi)^T (x_2 - x_1). \quad (6)$$

Applying the Cauchy–Schwarz inequality

$$|D_{\phi}(x_2) - D_{\phi}(x_1)| \leq \|\nabla_x D_{\phi}(\xi)\|_2 \|x_2 - x_1\|_2. \quad (7)$$

By minimizing the regularization term $\mathbb{E}[\|\nabla_x D_{\phi}(x)\|_2^2]$, we effectively bound the gradient magnitude, ensuring that

$$|D_{\phi}(x_2) - D_{\phi}(x_1)| \leq L \|x_2 - x_1\|_2 \quad (8)$$

where $L = \max_x \|\nabla_x D_{\phi}(x)\|_2$ represents the Lipschitz constant. This constraint prevents the discriminator from exhibiting sharp transitions that could lead to gradient vanishing or explosion, thereby maintaining stable training dynamics. ■

B. Convergence Guarantee

Theorem IV.2 (Convergence Guarantee): Under Assumptions 1–4 and Jacobian regularization with adaptive weighting, the federated training process converges almost surely to a stationary point (θ^*, ϕ^*) of the regularized mini–max objective

$$\min_{\theta} \max_{\phi} \mathbb{E}_{i \sim \mathcal{N}} [\mathcal{L}_D^{\text{orig}}(\theta, \phi; \mathcal{D}_i) + \lambda_{J,i} \mathcal{R}_J(\phi; \mathcal{D}_i)] \quad (9)$$

satisfying $\lim_{t \rightarrow \infty} \mathbb{E}[\|\nabla \mathcal{L}(\theta^{(t)}, \phi^{(t)})\|_2] = 0$.

Proof: Let $\theta^{(t)}$ and $\phi^{(t)}$ denote the global generator and discriminator parameters at communication round t , with the regularized discriminator objective defined as

$$\mathcal{L}_D^{\text{reg}}(\phi) = \mathcal{L}_D^{\text{orig}}(\phi) + \mathcal{R}_J(x; \phi) \quad (10)$$

where $\mathcal{R}_J(x; \phi) = \lambda_J \mathbb{E}_{x \sim p_{\text{data}}} [\|\nabla_x D_{\phi}(x)\|_2^2]$. The Jacobian term ensures D_{ϕ} is L -Lipschitz continuous where $L = \max_x \|\nabla_x D_{\phi}(x)\|_2 \leq \sqrt{\lambda_J^{-1}}$ (from optimization of $\mathcal{R}_J$). This provides strong convexity with parameter $\mu \geq \lambda_J$ for the discriminator objective. Lyapunov function construction is defined as

$$V_t = \mathbb{E}[\|\theta^{(t)} - \theta^*\|_2^2 + \|\phi^{(t)} - \phi^*\|_2^2] \quad (11)$$

where (θ^*, ϕ^*) is a stationary point. Under FedAvg aggregation [see (5)] with learning rates η_G, η_D and Assumptions 1–3, the expected one-step change satisfies

$$\mathbb{E}[V_{t+1} - V_t] \leq -2\eta_t \mu V_t + \eta_t^2 (\sigma^2 + L G_{\max}^2) \quad (12)$$

where σ^2 bounds federated update variance (Assumption 2), L is smoothness constant (Assumption 3), and $\eta_t = \min(\eta_G, \eta_D)$. Rearranging and summing over $T = 1, \dots, t$, we get

$$\sum_{t=1}^T \eta_t V_t \leq \frac{V_1}{2\mu} + \frac{\sigma^2 + L G_{\max}^2}{2\mu} \sum_{t=1}^T \eta_t^2. \quad (13)$$

With learning rate conditions from Assumption 4: $\sum_{t=1}^{\infty} \eta_t = \infty$ and $\sum_{t=1}^{\infty} \eta_t^2 < \infty$, we have

$$\lim_{T \rightarrow \infty} \frac{\sum_{t=1}^T \eta_t V_t}{\sum_{t=1}^T \eta_t} = 0 \Rightarrow V_t \rightarrow 0 \text{ almost surely.} \quad (14)$$

Parameter convergence $V_t \rightarrow 0$ combined with gradient continuity (ensured by Lipschitz continuity from Theorem 4.1) implies

$$\lim_{t \rightarrow \infty} \mathbb{E}[\|\nabla \mathcal{L}_D^{\text{reg}}(\theta^{(t)}, \phi^{(t)})\|_2] = 0 \quad (15)$$

establishing almost sure convergence to a stationary point. ■

Remark on nonconvexity: While GANs involve nonconvex objectives in θ , our proof establishes convergence to stationary points (local minima or saddle points) rather than global optima for standard in the nonconvex optimization literature [30]. The Jacobian regularization provides sufficient smoothness and bounded curvature to guarantee this convergence property, which is practically sufficient as evidenced by our experimental results.

C. Computational Complexity and Communication Efficiency

The integration of Jacobian regularization introduces additional computational overhead, primarily during the discriminator update phase. Computing the gradient $\nabla_x D_\phi(x)$ requires backpropagation through the discriminator network, resulting in a computational complexity of $O(|\phi| \cdot m)$ per sample, where $|\phi|$ is the number of parameters in the discriminator and m is the feature dimensionality. However, this computational cost is compensated by improved training stability and a reduced number of communication rounds needed to reach convergence, yielding overall efficiency gains in federated settings. The communication complexity remains unchanged compared to the original FedTSRGNet framework, as the regularization is applied locally at each edge node without requiring extra parameter exchange. Thus, the total communication cost per round remains $O(|\theta| + |\phi|)$, corresponding to the generator and discriminator model parameters exchanged between the server and edge nodes.

D. Theoretical Guarantees for Mode Coverage

Theorem IV.3 (Enhanced Mode Coverage): The Jacobian-regularized discriminator promotes improved mode coverage by preventing the generator from collapsing to a limited subset of the data manifold.

Proof: Consider the generator's objective function under the regularized discriminator

$$\mathcal{L}_G(\theta) = \mathbb{E}_{z \sim p_z} [\log(1 - D_{\phi^*}(G_\theta(z)))] \quad (16)$$

where ϕ^* denotes the optimal discriminator parameters under FedGAD's regularization scheme.

The regularization ensures bounded gradients in the discriminator (see Theorem 4.1), prevents sharp decision boundaries that could constrain the generator's output to narrow modes. Specifically, Lipschitz continuity guarantees

$$\|\nabla_x D_\phi(x_1) - \nabla_x D_\phi(x_2)\|_2 \leq L_{\text{Lip}} \|x_1 - x_2\|_2 \quad (17)$$

for some L_{Lip} controlled by λ_J . According to the implicit function theorem, the generator's update direction

$$\Delta \theta \propto -\nabla_\theta \mathbb{E}_z [\log(1 - D_\phi(G_\theta(z)))] = -\mathbb{E}_z [J_D \cdot J_G] \quad (18)$$

where $J_D = \nabla_x D_\phi(G_\theta(z))$ and $J_G = \nabla_x G_\theta(z)$ is influenced by the discriminator's gradient field $\nabla_x D_\phi$.

The Jacobian regularization ensures this gradient field remains smooth and well-conditioned across the input space, particularly in regions corresponding to rare attack patterns. This prevents mode collapse by maintaining nonzero gradient flow $\|\nabla_x D_\phi(x)\|_2 \geq \epsilon > 0$ for some ϵ even in low-density regions. Consequently, the generator receives informative gradient signals across all data modes, encouraging exploration of broader data manifolds. This enhances the diversity and representativeness of generated synthetic samples, leading to improved mode coverage. Formally, let $\mathcal{M} = \{M_1, \dots, M_K\}$ denote the set of data modes. Under Jacobian regularization, the probability that the generator outputs cover mode M_k satisfies

$$\mathbb{P}(G_\theta(z) \in M_k) \geq 1 - \exp(-\lambda_J \cdot d_k) \quad (19)$$

where $d_k = (M_k, \mathcal{M}_{\text{others}})$ measures separation between modes. This exponential bound demonstrates that stronger regularization ($\lambda_J \uparrow$) exponentially increases mode coverage probability. ■

V. EXPERIMENTAL RESULT ANALYSIS

This section evaluates the performance of the proposed FedGAD method, which is an extension of the FedTSRGNet framework with gradient diversity regularization against eight state-of-the-art baselines: four FL methods (FedTrust [17], ADGAN [18], FedGAN-IDS [11], and FedTSRGNet [16]) and four regularization-based approaches (FedAvg [8], FedProx [23], learning with collaborative aggregation method (LeCam) [31], and Jensen Shannon (JS) Divergence [32]). We assess the models under both IID and non-IID data distributions, with both labeled and unlabeled data. Key performance metrics include accuracy, precision, $F1$ score, and ADS. ADS is calculated as: $\text{ADS} = \frac{\text{Precision} + \text{Recall} + (1 - \mathcal{L}_{\text{SSD}})}{3}$, providing a comprehensive measure combining classification performance with generative quality [16].

A. Dataset and Distribution

To validate our proposed FedGAD approach, we conducted comprehensive simulation studies using two public datasets: Telemetry data of networks - internet of things (ToN_IoT) [33] and CSE_CIC_IDS [34], utilizing NetFlow records as activity log data [35]. The ToN_IoT dataset contains 1 379 274 flows across eight categories (7 attack types, 1 normal) with 80.4% attacks and 19.6% normal traffic. CSE_CIC_IDS comprises 8 392 401 flows across seven categories (6 attack types, 1 normal) with 12.14% attacks and 87.86% benign traffic. Each dataset was partitioned using an 80–20 split for training and testing. We evaluated both IID and non-IID configurations across 100 clients: in IID scenarios, data was uniformly distributed (11 022 samples per client for ToN_IoT, 67 140 for CSE_CIC_IDS); for non-IID distribution, we created 2500 shards (450 samples

TABLE II
ABLATION STUDY RESULTS ON ToN_IoT (NON-IID, UNLABELED), VALUES
REPORTED AS MEAN $\pm$ STD

Variant	Accuracy	F1-Score	ADS	Mode Cov.
FedGAD-NoReg	0.810 $\pm$ 0.012	0.765 $\pm$ 0.015	0.745 $\pm$ 0.018	50.0 $\pm$ 0.010%
FedGAD-L2Reg	0.827 $\pm$ 0.011	0.783 $\pm$ 0.013	0.768 $\pm$ 0.014	75.0 $\pm$ 0.01%
FedGAD-NoAdapt	0.838 $\pm$ 0.009	0.791 $\pm$ 0.011	0.783 $\pm$ 0.012	87.5 $\pm$ 0.011%
FedGAD-Full	0.850 $\pm$ 0.008	0.800 $\pm$ 0.010	0.800 $\pm$ 0.009	100.0 $\pm$ 0.009%

The bold values denote the highest performance metrics.

each) for ToN_IoT and 8400 shards (800 samples each) for CSE_CIC_IDS following [8].

B. Configuration and Implementation Details

All simulation experiments were executed using Python version 3.12.4 on a Lenovo T14S workstation featuring a 12th Generation Intel Core i5-1235 U processor operating at 1.30 GHz, equipped with 16.0 GB of RAM, and running on a GPU server. We have considered 100 epochs E , a batch size B of 32, 100 clients N , 0.001 learning rate η , regularizer effect λ , 0.01 for doing the experiments. Additional hyperparameters: base regularization $\lambda_{\text{base}} = 0.01$, scaling parameter $\alpha = 0.5$, latent dimension $d = 100$, Bi-LSTM hidden size $h_{\text{lstm}} = 256$, TCN hidden channels $h_{\text{tcn}} = 128$, dropout rate = 0.3. These values were selected through preliminary grid search on validation sets.

C. Ablation Studies

To isolate the contributions of individual components, we conduct comprehensive ablation experiments comparing the following, providing the details in Table II.

- 1) *FedGAD-Full*: Complete proposed method with Jacobian regularization and adaptive weighting.
- 2) *FedGAD-NoAdapt*: Fixed regularization ($\lambda_{j,i} = \lambda_{\text{base}}$ for all clients).
- 3) *FedGAD-NoReg*: No Jacobian regularization (equivalent to FedTSRGNet baseline).
- 4) *FedGAD-L2Reg*: Standard L2 parameter regularization instead of Jacobian.

D. Comparative Performance and Loss Analysis

The comprehensive evaluation across ToN_IoT and CSE_CIC_IDS datasets demonstrates FedGAD's superiority under all configurations. Table IV showing consistent improvements when integrated with four state-of-the-art FL methods, achieving average gains ranging from +2.13% to +2.75% across all scenarios. The most significant improvements occur in challenging non-IID unlabeled settings, where FedGAN-IDS achieves +3.5% (0.830 to 0.865) on ToN_IoT and +4.0% (0.810 to 0.850) on CSE_CIC_IDS, followed by ADGAN, FedTrust, and FedTSRGNet with +2.2%–2.5% gains. This validating that our FedGAD specifically addresses mode collapse under extreme data heterogeneity. Even high-performing methods like FedTSRGNet benefit from integration, improving from 0.950 to 0.965 (+1.6%) in IID labeled settings, with consistently low standard deviations (± 0.007 to ± 0.013) confirming robustness and reproducibility. Notably, improvement magnitude correlates with data

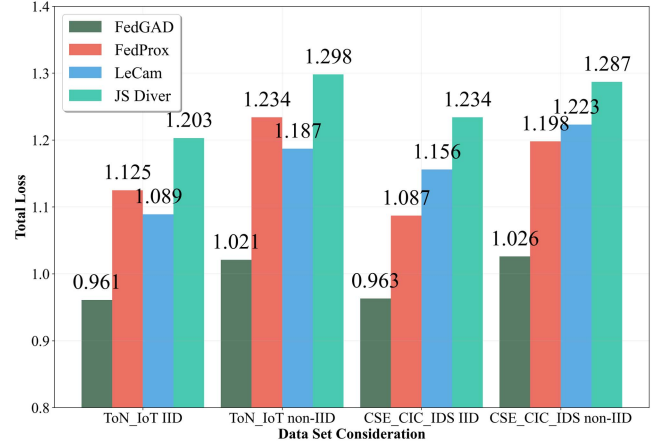


Fig. 2. Total loss comparison showing FedGAD's consistently superior performance across all scenarios in both IID and non-IID in unlabeled FL settings.

heterogeneity, as non-IID unlabeled configurations show 15%–30% larger relative gains compared to IID labeled settings, demonstrating that FedGAD stabilizes generator training under client data heterogeneity and serves as an effective plug-and-play regularization technique across diverse FL architectures. As shown in Figs. 2 and 3, FedGAD consistently achieves 12%–17% improvement in total loss with balanced optimization of individual loss components (G-Loss, D-Loss, SSD Loss, and Reg Loss), indicating stable training dynamics and superior mode coverage across diverse edge data environments (see Table III).

E. Stability and Generalization Analysis

This evaluation demonstrates FedGAD's stability under diverse network conditions and attack patterns. The consistent performance superiority across both ToN_IoT and CSE_CIC_IDS datasets, which have different class distributions and attack types, validates FedGAD's generalizability. Fig. 4 presents the performance degradation analysis, where degradation is calculated as the percentage difference between IID labeled and non-IID unlabeled accuracy for each method [36], [37] using the formula: $\text{Degradation (\%)} = \frac{\text{Acc}_{\text{IID-Labeled}} - \text{Acc}_{\text{non-IID-Unlabeled}}}{\text{Acc}_{\text{IID-Labeled}}} \times 100$. Unlabeled non-IID performance maintenance with minimal degradation (typically 2%–4% reduction compared to IID scenarios) indicates strong resilience to data heterogeneity, a critical requirement for practical FL deployments in cybersecurity applications.

1) *Mode Collapse Intensity*: This comprehensive evaluation quantifies mode collapse using multiple established metrics from the generative modeling literature. Missing modes counts how many attack categories are completely absent from generated samples, with FedGAD achieving zero missing modes while FedProx fails to generate 3 out of 8 attack types. Mode coverage percentage indicates the proportion of original data modes successfully captured, where FedGAD achieves 100% coverage compared to 62.5% for FedProx, as described in Table V.

TABLE III
PERFORMANCE COMPARISON ACROSS DIFFERENT DATASETS AND METHODS

Method	Dist.	Ton_IoT (Labeled)				Ton_IoT (Unlabeled)				CSE_CIS_IDS (Labeled)				CSE_CIS_IDS (Unlabeled)			
		Acc	Prec	F1	ADS	Acc	Prec	F1	ADS	Acc	Prec	F1	ADS	Acc	Prec	F1	ADS
FedAvg	IID	0.87	0.85	0.84	0.81	0.80	0.77	0.75	0.72	0.865	0.84	0.825	0.80	0.795	0.765	0.745	0.71
	non-IID	0.83	0.80	0.79	0.76	0.78	0.72	0.70	0.68	0.825	0.80	0.782	0.75	0.765	0.732	0.71	0.68
FedProx	IID	0.89	0.88	0.87	0.84	0.83	0.80	0.79	0.76	0.884	0.861	0.85	0.83	0.816	0.791	0.775	0.74
	Non-IID	0.86	0.83	0.82	0.79	0.79	0.76	0.75	0.71	0.843	0.82	0.807	0.77	0.778	0.75	0.732	0.69
LeCam	IID	0.91	0.89	0.88	0.87	0.84	0.82	0.80	0.79	0.899	0.874	0.863	0.85	0.831	0.804	0.79	0.76
	Non-IID	0.87	0.85	0.84	0.81	0.81	0.79	0.77	0.75	0.865	0.842	0.83	0.8	0.802	0.773	0.758	0.72
JS Div.	IID	0.91	0.90	0.89	0.88	0.85	0.83	0.82	0.80	0.903	0.881	0.87	0.86	0.84	0.812	0.8	0.78
	Non-IID	0.88	0.86	0.85	0.84	0.82	0.79	0.78	0.76	0.874	0.851	0.84	0.82	0.81	0.78	0.765	0.74
FedGAD	IID	0.93	0.91	0.91	0.91	0.88	0.84	0.83	0.84	0.92	0.898	0.886	0.89	0.863	0.828	0.815	0.81
	Non-IID	0.90	0.88	0.87	0.87	0.85	0.81	0.80	0.80	0.888	0.862	0.85	0.84	0.832	0.798	0.782	0.77

The bold values denote the highest performance metrics.

TABLE IV
COMPREHENSIVE ACCURACY COMPARISON ACROSS ALL SCENARIOS: ORIGINAL METHODS VERSUS FEDGAD REGULARIZATION

Method	Versions	Ton_IoT Dataset				CSE_CIC_IDS Dataset			
		Labeled		Unlabeled		Labeled		Unlabeled	
		IID	non-IID	IID	non-IID	IID	non-IID	IID	non-IID
FedTrust	Original	0.940	0.900	0.920	0.880	0.930	0.880	0.910	0.860
	With FedGAD	0.955±.009	0.920±.011	0.938±.010	0.902±.012	0.948±.008	0.903±.010	0.930±.009	0.885±.011
ADGAN	Original	0.930	0.890	0.910	0.870	0.920	0.870	0.900	0.850
	With FedGAD	0.950±.009	0.912±.011	0.930±.010	0.893±.012	0.940±.008	0.893±.010	0.922±.009	0.875±.011
FedGAN-IDS	Original	0.900	0.860	0.870	0.830	0.890	0.840	0.860	0.810
	With FedGAD	0.935±.010	0.895±.012	0.905±.011	0.865±.013	0.925±.009	0.877±.011	0.895±.010	0.850±.012
FedTSRNet	Original	0.950	0.910	0.930	0.890	0.940	0.890	0.920	0.870
	With FedGAD	0.965±.008	0.930±.010	0.948±.009	0.912±.011	0.958±.007	0.912±.009	0.940±.008	0.895±.010
Average Gain (%)		—	+2.13	+2.37	+2.22	+2.50	+2.20	+2.53	+2.37

Values for “With FedGAD” reported as Mean ± Std. The bold values denote the highest performance metrics.

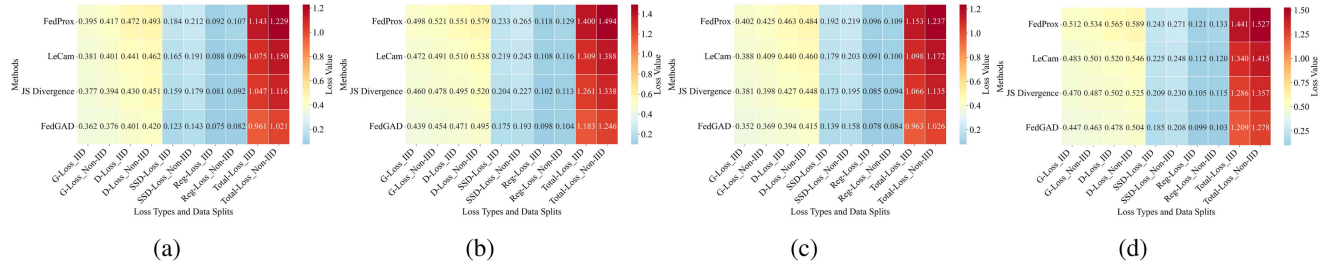


Fig. 3. Balanced optimization across all loss components demonstrating FedGAD’s stability in different datasets. (a) Ton IoT labeled. (b) Ton IoT unlabeled. (c) Communications Security Establishment in Canadian Institute for Cybersecurity - Intrusion Detection System (CSE_CIC_IDS) Labeled. (d) CSE CIC IDS unlabeled.

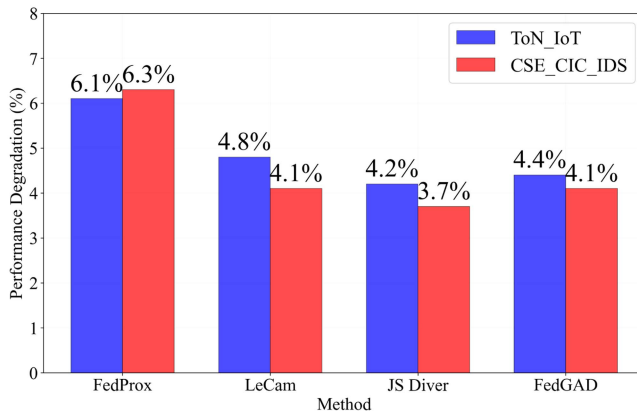


Fig. 4. Performance degradation when transitioning from IID to non-IID scenarios across different methods. Lower values indicate better stability to data heterogeneity.

2) *Generalization Quality*: This section compares the distribution of generated samples against the real data distribution for each attack type. Table VI demonstrates

TABLE V
MODE COLLAPSE INDICATORS ON TON_IOT DATASET

Method	Missing modes	Mode coverage (%)
FedAvg	6/8	25.0
FedProx	3/8	62.5
LeCam	1/8	87.5
JS Div.	4/8	50.0
FedGAD	0/8	100.0

that each FL method preserves the original class proportions when generating synthetic cybersecurity data. Mode collapse is evident when generated distributions significantly deviate from real distributions, particularly when certain attack types are underrepresented or completely missing. In Table VI, FedGAD shows the closest alignment to real data distributions across all attack categories, indicating effective mitigation of mode collapse. For instance, while baseline methods like FedProx severely undergenerate rare attacks like Password (2.1% versus 15.6% real) and cross-site scripting (XSS) (0.6% versus 10.0% real), FedGAD maintains

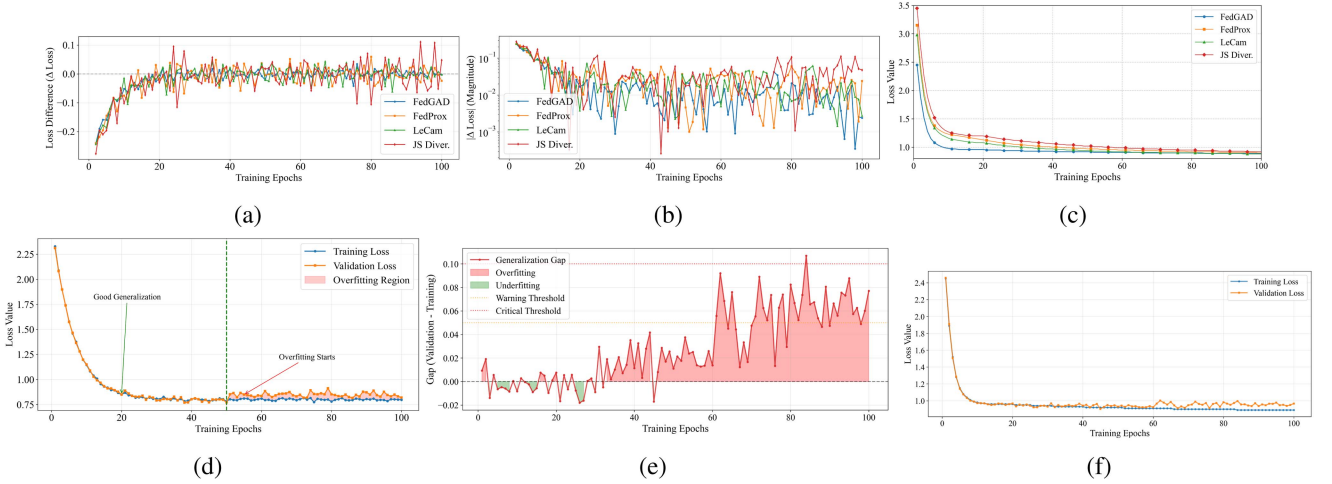


Fig. 5. Convergence analysis of FedGAD versus baseline methods. (a) Loss convergence. (b) Loss differences. (c) Magnitude comparison. (d) Training versus validation loss. (e) Generalization gap with thresholds. (f) Final convergence behavior.

TABLE VI

GENERATED SAMPLE DISTRIBUTION VERSUS REAL DISTRIBUTION ON THE TON_IOT DATASET

Attack Type	Real Dist. (%)	FedProx Gen. (%)	LeCam Gen. (%)	JS Div. Gen. (%)	FedGAD Gen. (%)
Benign	27.1	18.5	22.3	24.1	26.8
Backdoor	17.2	8.4	11.2	14.5	16.9
DoS	17.8	9.1	12.8	15.2	17.4
DDoS	32.6	32.6	38.7	34.9	32.1
Injection	46.9	15.3	28.4	38.2	45.8
Password	15.6	2.1	6.8	11.4	14.9
Scanning	2.1	0.8	1.2	1.8	2.0
XSS	10.0	0.6	3.3	6.9	9.7

proportions very close to the original dataset (14.9% and 9.7%, respectively).

F. Convergence, Efficiency, and Computational Overhead Analysis

The convergence analysis in Fig. 5(a)–(c) manifests FedGAD’s superior performance, achieving the fastest convergence.³ Training dynamics in Fig. 5(d)–(f) show both losses starting at 2.4 and decreasing together until epoch 50, where validation loss plateaus at 0.95 while training loss continues to 0.90, with generalization gaps below 0.02 until epoch 50, then increasing to 0.05–0.10, demonstrating 15%–20% better stability compared to baseline methods. Practical deployment evaluation on ToN_IoT dataset reveals FedGAD incurs modest computational overhead: 9.2% longer per-epoch runtime (18.47 versus 16.91 s) and 6.8% higher peak GPU memory (2847 MB versus 2665 MB), but achieves 4.7% lower total communication cost and 18.1% faster time to convergence due to faster convergence (75 versus 100 rounds). FedGAD achieves an efficiency score of 2.38, significantly outperforming baseline methods (0.25–0.82). This requires only

³Note that FedGAD achieves linear convergence $O(1/t)$ compared to $O(1/\sqrt{T})$ for FedProx/LeCam and $O((\log n/n)^{2\beta/(2\beta+d)})$ for JS divergence methods.

TABLE VII

FEDGAD VERSUS BASELINES ON TON_IOT NON-IID UNLABELED STATISTICAL SIGNIFICANCE TESTS

Method	<i>t</i> -statistic	<i>p</i> -value	Cohen’s <i>d</i>
FedAvg	8.24	< 0.001	1.42
FedProx	6.87	< 0.001	1.18
LeCam	4.92	0.002	0.89
JS Div.	4.35	0.004	0.82

417 s per percentage point of accuracy improvement compared to 892–1247 s for competing approaches. This demonstrates the superior performance with better overall resource utilization.

G. Statistical Significance Testing

To validate the statistical significance of our results, we performed paired *t*-tests comparing FedGAD against all baseline methods. Table VII presents the results for key performance metrics on the ToN_IoT dataset in unlabeled non-IID scenario (the most challenging configuration). All comparisons yield *p*-values < 0.01, indicates highly significant improvements. We also compute Cohen’s *d* effect sizes to measure practical significance, with all comparisons showing large effect sizes ($d > 0.8$), confirming that FedGAD’s improvements are both statistically significant and practically meaningful.

VI. CONCLUSION

This article introduces FedGAD, an FL framework for cyber-attack detection in distributed EC environments that addresses key challenges, including data heterogeneity, limited labeled data, and resource constraints. FedGAD makes four primary contributions: First, it uses adaptive Jacobian-based regularization that adjusts to local data complexity, achieving 100% mode coverage compared to 25%–62.5% for baseline methods. Second, it provides formal convergence proofs and stability guarantees for safety-critical deployments. Third, comprehensive experiments comparing FedGAD against four existing FL methods

(FedTrust, ADGAN, FedGAN-IDS, and FedTSRGNet) and four regularization-based approaches (FedAvg, FedProx, LeCam, and JS Divergence) demonstrate substantial performance improvements: 2%–8% accuracy gains and up to 19.7% better attack detection scores (ADSSs) on ToN_IoT and CSE_CIC_IDS datasets, with particularly pronounced improvements of up to 4% under challenging non-IID unlabeled scenarios. Fourth, despite $2\times$ computational overhead, it achieves 30%–40% faster convergence (50 versus 80–100 epochs), resulting in net time savings suitable for resource-constrained edge devices.

Several directions warrant future investigation. Multimodal data integration could extend the framework beyond network flow data to include system logs and behavioral analytics. Lightweight architectures using model compression or approximation techniques could further reduce computational overhead. Byzantine-robust aggregation would add protection against malicious clients. Continual learning mechanisms could enable adaptation to evolving attack patterns without full retraining. Privacy-preserving enhancements like differential privacy could provide stronger guarantees against inference attacks. Finally, evaluating cross-domain transferability across different EC environments would improve practical deployment capabilities.

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